

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8510**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00 am - 11:00 am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600056
Course Section:	1				

L_29_513_986_1_1505_38310



Name of the candidate. Anushka Shondge

Candidate's UID number:

2	0	2	2	6	0	0	0	5	6
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Examination class room number:

4	0	1
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Invigilator's signature with date:

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(To be filled in by the candidate)
EXAMINATION: BE-MSE B.Tech
(For example FY MCA/FY BTECH)
PROGRAM: mse AIML
DAY AND DATE: 8/10/25 & 20th Tuesday
COURSE NAME: DL

SEMESTER : 7
I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



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(Begin answer for each question on a new page)

Q1

$$a) f(w) = \frac{1}{2} (xw - y)^2$$

~~$$\frac{df}{dw} = \frac{1}{2} \times (xw - y) \times x$$

$$= (xw - y) \times x$$

$$= x(xw - y)$$~~

~~$$\frac{df}{dx} = \frac{1}{2} \times (xw - y)^2 \times w = w(xw - y)$$~~

~~$$\frac{df}{dy} = \frac{1}{2} \times (xw - y) \times -1 = -(xw - y)$$

$$= -w(xw - y)$$~~

~~$$\frac{df}{dx} = \frac{w(xw - y)}{-1(x - y)}$$~~

Update rule of gradient descent

$$x_{k+1} = x_k - \alpha \nabla f(x_k)$$

Here α = learning rate

x_k = current value of x .

x_{k+1} = Next value of x

$\nabla f(x)$ = Loss function.



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Question
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$$\left[\frac{w(xw-y)}{-(xw-y)} \right] \downarrow$$

~~fraction~~ $do =$

$$f(w) = \left[\frac{w(xw-y)}{-(xw-y)} \right]$$

$H =$

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Q2

b) ① Autoencoder is used to build the latent representation from the input image and then reconstruct the input output image from that latent version.

② Encoder Eq: Build the latent version

$$E = \phi(w_x \times x + b)$$

③ Decoder Eq: Reconstruct the output image

$$D = \phi(w_x \times E + b)$$

$$E \rightarrow MSE = \frac{1}{n} \sum (x - \hat{x})^2$$

$\downarrow$ $\searrow$
 input output
 image image.

④ Under complete ~~auto~~ auto encoder the ~~2~~ ~~no~~ neurons in the ~~2~~ hidden layer are less than that of input layer

⑤ So the only essential and important features are ~~rest~~ represented through the latent version.

⑥ ~~Thus~~ This causes image compression or dimensionality reduction.

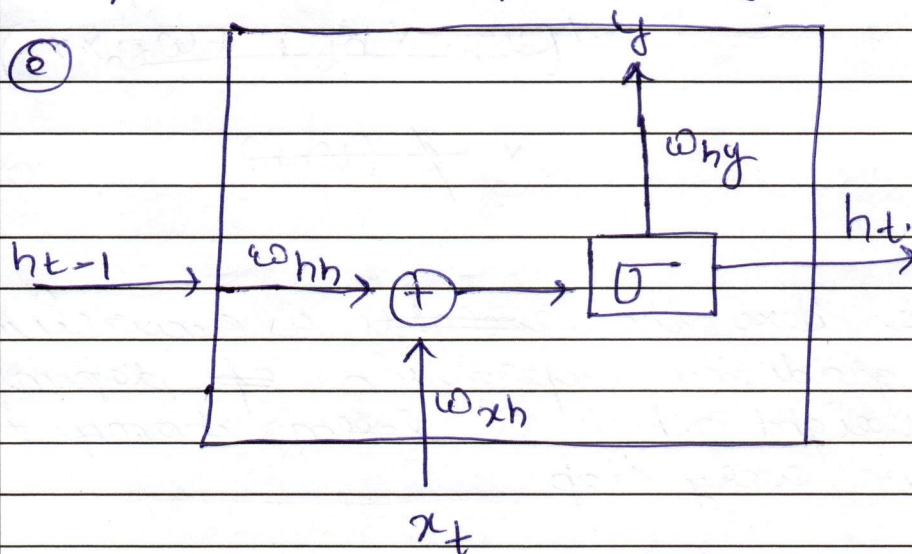
Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
		① PCA is also a dimensionality reduction technique for linear classes.
		② So linear undercomplete autoencoder trained with MSE learns subspace equivalent to PCA.
		③ In
		④ Overcomplete autoencoder the neurons in the hidden layer are more than or equal to that of input layer.
		⑤ So it has full potential to just replicate the exactly the input image and even unnecessary patterns and features.
		⑥ This may lead to trivial identity mapping of input to output and this is not generalized model as it just memorises the input image without understanding the underlying patterns.
		⑦ So regularization of these image is necessary to get the useful interpretation of the image.
		⑧ There are 2 major regularization techniques which are used
		① Sparse autoencoder
		② Denoising autoencoder.

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Q3

a) ① RNN is the sequence model. It can take input where order of input is important for predicting the output.



This is the architecture of the RNN
The equation of hidden state

$$h_t = \phi(w_{hh} \times h_{t-1} + w_{xh} \times x_t)$$

Here, $h_t \rightarrow$ current hidden state

$h_{t-1} \rightarrow$ previous hidden state

$w_{hy}, w_{hh}, w_{xh} \rightarrow$ weights

$x_t \rightarrow$ Input

$y \rightarrow$ Output.

Equation of output state

$$y = \phi(w_{hy} \times h_t)$$

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Recursive Gradient expression,

$$\frac{dh_t}{dh_{t-1}} = \phi'(w_{hh}) w_{hh}$$

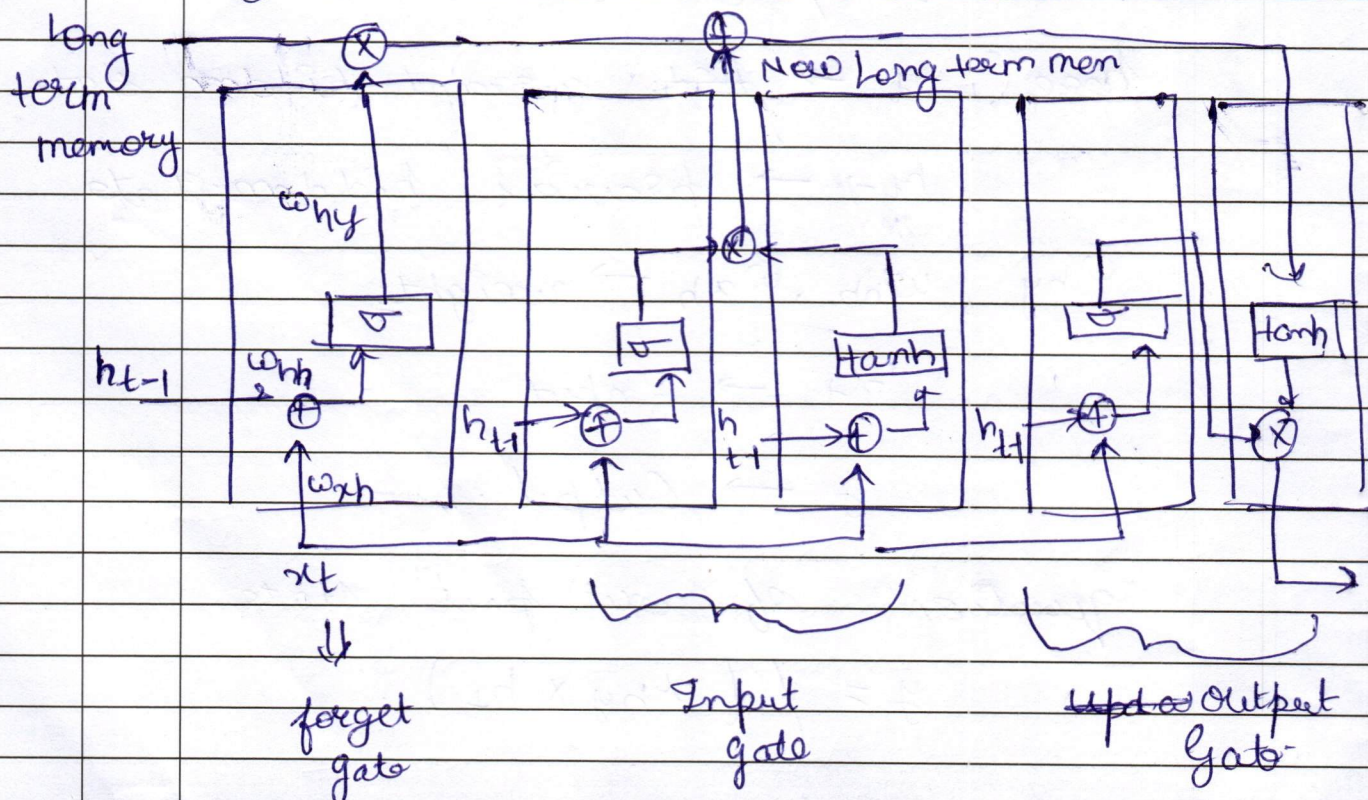
$$\phi'(w_{hh} \times h_{t-1} + w_{xh} \times x_t)$$

$$\times \phi'(w_{hh})$$

③ Here the recursive gradient expression depends on weight which is learn changing in every step

④ So it will lead to every small value which may cause loss of vanishing gradient problem

⑤ To overcome this there is one more model. LSTM It carries 2 different state: hidden state and the cell state



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$$f = \sigma(w_{hh} h_{t-1} + w_{xh} x_t)$$

→ forget gate

$$i = \sigma(w_{hh} h_{t-1} + w_{xh} x_t)$$

$$c = \tanh(w_{hh} h_{t-1} + w_{xh} x_t)$$

New long term memory = $f \times h_{t-1} + i \times c$

$$o = \sigma(w_{hh} h_{t-1} + w_{xh} x_t)$$

New short term memory = $o \times \tanh(\text{New long term mem})$

Here recursive gradient expression

$$\frac{dc}{dh_{t-1}} = f$$

Hence ~~forget~~ f is forget gate value

which is passed to by sigmoid function which will give from 0 to 1 and ~~it~~ f is always

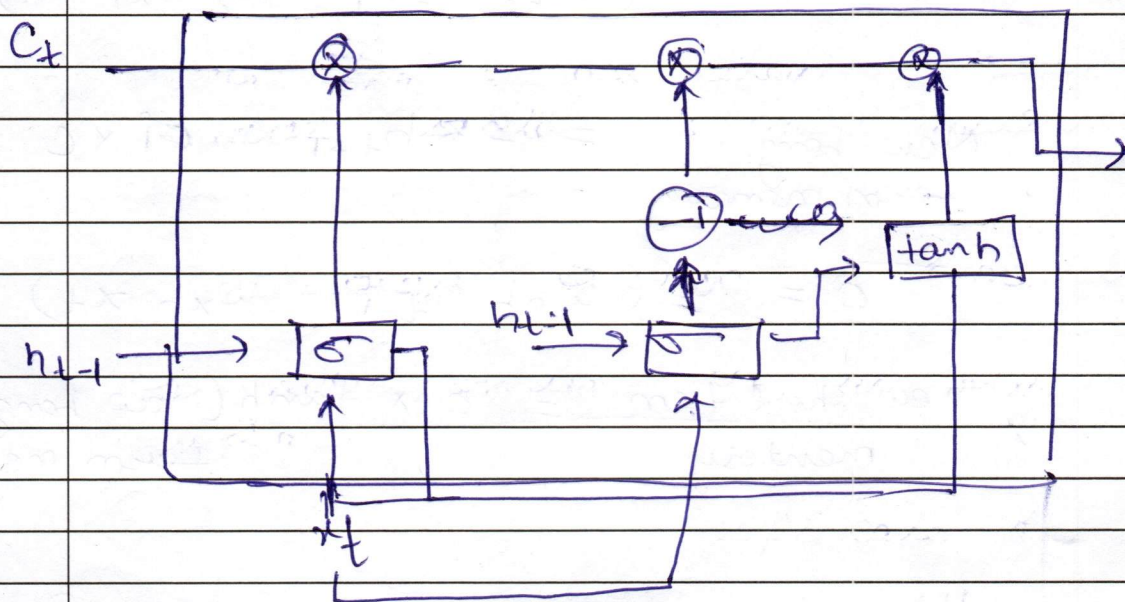
so there is no vanishing gradient problem.

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Q3

b) GRU is optimized version of LSTM.
It has fewer parameters as compared to LSTM



This is the architecture of GRU
It has only 2 gate and only 1 state

$$r = \sigma(w_{hh} h_{t-1} + w_{xh} x_t)$$

$$c = \tanh(r \times (h_{t-1} \times x_t))$$

$$u = \sigma(w_{hh} h_{t-1} + w_{xh} x_t)$$

① The reset gate check that how much of past information is relevant to current

② If $u \neq 0$ the reset gate passes the output to candidate gate potential

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memory

③ Update gate is main deciding factor to keep the combination of new and old information ~~multi~~

1st m Equation consist of
3 gate and 2 ~~hidden~~ state

we need to 2 state i.e hidden state and cell state.

Here we are using only 1 hidden state throughout to store long term as well short & ~~the~~ term memory.

$$C_t = f \times h_{t-1} + i \times C$$

This is the final cell state of LSTM.
and here the hidden state is
itself is the cell state for
storing and calculating.

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Q1

b) ① Overfitting occurs when the ~~mod~~ hidden state has more neuron than the input state

② This occurs because ~~we~~ model memorize the value rather generalizing or understanding the underlying patterns.

③ ~~It~~ It tries to replicate the exact input to output

④ ~~→~~ This is observed in overcomplete autoencoder where without regularization the ~~no~~ model's output is useless.

~~Q1~~ ⑤ So Overfitting occurs in such models as ~~It~~ it does not learn from the ~~by~~ by extracting features.

[illegible]

[illegible]

[illegible]

[illegible]